LESSON PLAN

AIRCRAFT PROPULSION & ENGINEERING SYSTEM (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	14	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Corrosion treatment of engine components
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Corrosion treatment of engine components.
5. Specific Objectives	6.1.3.1 Trainee will be able to perform rectification of corrosion.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of corrosion removal using pickling paste and wire brushing on simulated engine component samples; Safety precautions include wearing gloves, goggles, and ensuring ventilation; Instructor supervision required during chemical handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of corrosion in aircraft engines?
(b) List three common types of corrosion found on engine components.
(c) Why is immediate rectification of corrosion critical for engine safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Identification of corrosion types on engine parts • Preparation of surface for corrosion removal • Application of pickling paste technique • Mechanical removal using wire brushes • Inspection of treated surfaces for defects	L & Ds	PPT, IB	15 min
(b)	• Neutralization and cleaning of treated areas • Application of protective coatings • Final inspection and documentation • Safety protocols during chemical use • Proper disposal of waste materials	L & Ds	PPT, PDF	20 min
(c)	• Review of rectification procedures • Common errors in corrosion treatment • Maintenance schedules for corrosion prevention • Reporting requirements for corrosion findings • Integration with overall engine maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the step-by-step process for rectifying corrosion on an engine component.
(b) Describe the safety measures required when handling pickling paste.
(c) How do you verify that corrosion has been completely removed?

12. Home Assignment and Follow Up	Prepare a brief report on the contents and applicability of the AHCP kit as covered in the next lesson plan (LP 10).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to AHCP kit, contents of kit and their applicability on aircraft.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min